Mobile Application Development

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Smester Project Documentation

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1. Introduction

Our smester project for mobile application development is a plant sharing app called ***Plantae*** app. It aims to create a platform for plant lovers to group together and share their plants and experience with related community. It will group toghether people who love plants and flowers. It help them to find people of similar interest – with planting hobby.

1.1 Features

It has following features:

- User can login.
- User can sign up.
- User has his/her profile.
- User can manage and edit profile.
- People can post about their plants , flowers and cuttings.
- People can post add about their plants and can label post as: *Sell*, *Give away*, *swap* (to exchange plants).
- Posts will circulate among all users of app on main page.
- Users can interact with eachother via comments.
- GPS Location of user is stored to see where plant is.
- People can see posts by filtering like if they want to see only posts in which plants can be exchanged, swap filter can be applied.
- Responsive mobile app.
- Need internet for proper working e.g. to get gps location.
- Log out functionality.

1.2 Tools and Technologies used

Following tools and Technologies has been used to develop the ***Plantae*** app:

- VS code – IDE for development.
- Flutter dart - programming language.
- Firebase – Baas (backen as a service).
- Firebase auth – User authentication.
- Cloud firestore – for storing data of users and posts.

1.3 Dart packages

Following packages are used in complete development:

- Geolocator – for GPs funcitoanlity.
- Google maps flutter.
- Permission_handler.
- Image_picker.
- Geocoding.
- Permission handler.

1.4 Designed figma for app

Following are the snapshots for figma designed for UI of app:

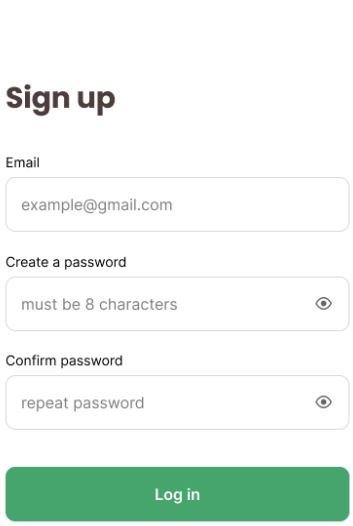

Figure 1: Sign Up

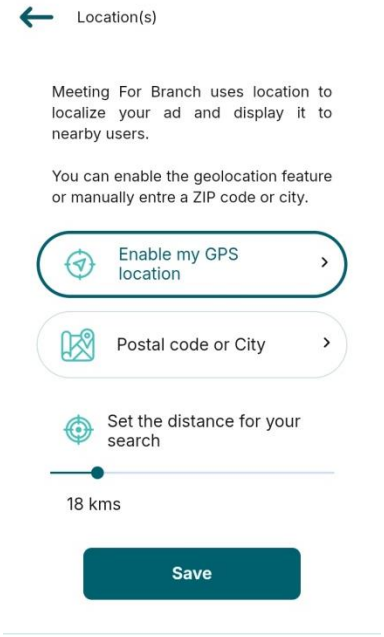

Figure 2: Get GPS location

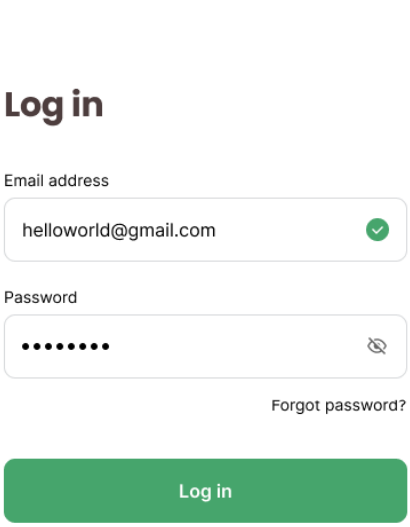

Figure 3: Log in Page

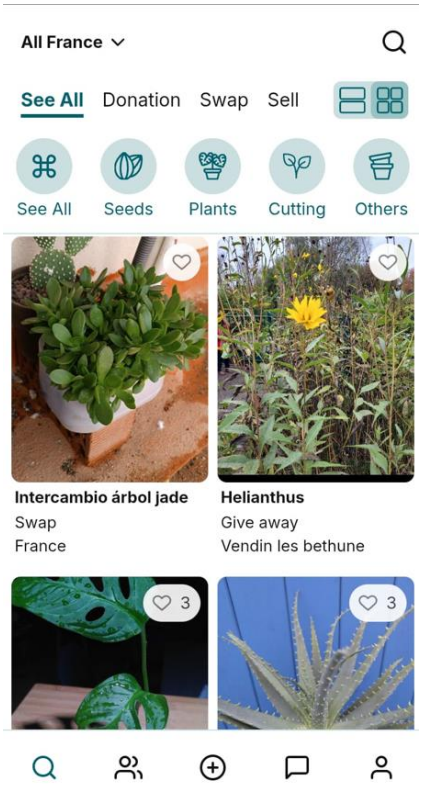

Figure 4: Home screen for new posts

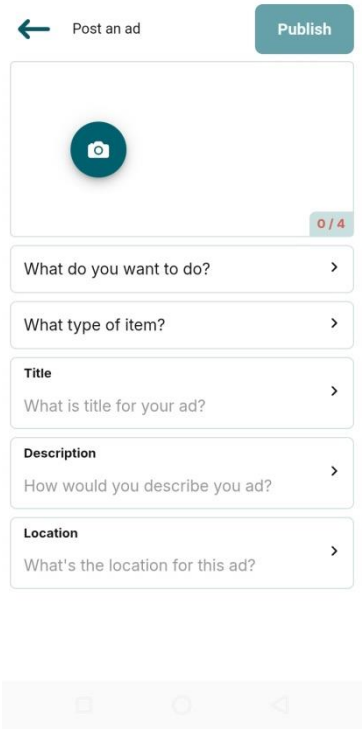

Figure 5: Add a Post

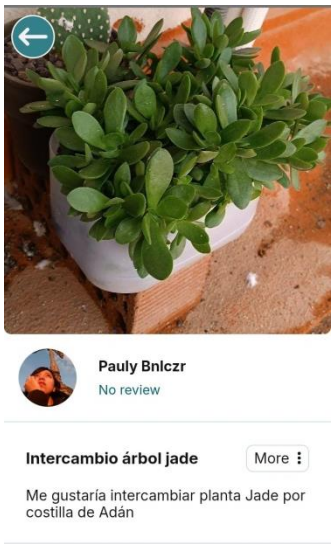

Figure 6: Post detail page

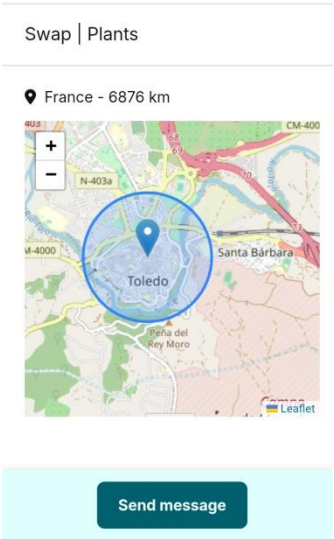

Figure 7: Location Map of post

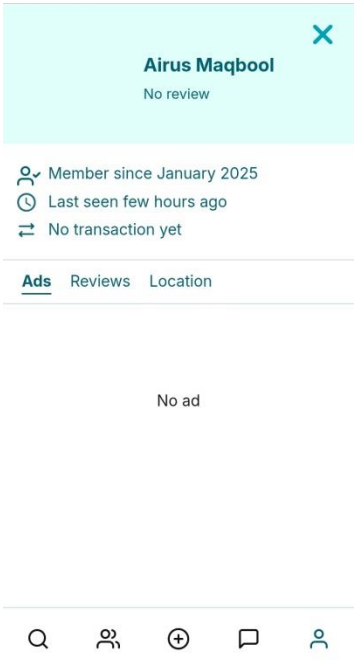

Figure 8: User profile

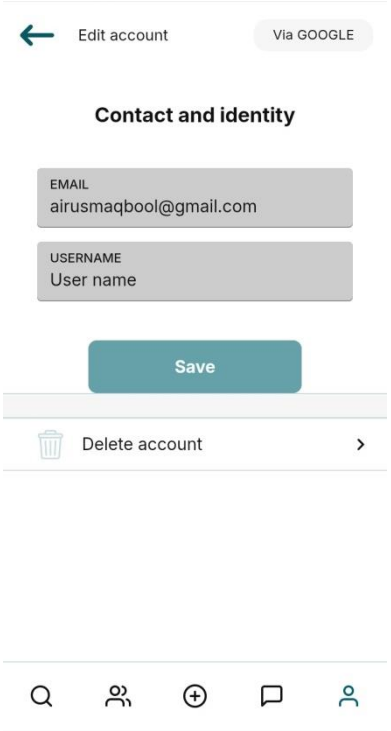

Figure 9: Mange user profile

Figure 10: Chats of app users.

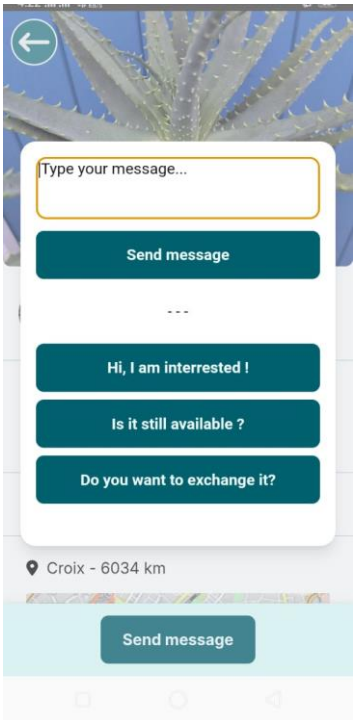

Figure 11: Messages of users.

1.5 Developed Pantae app snapshots

Following are the snapshots of developed app , screencshots are taken after installing app on mobile:

- **Sign up page:**
  User will sign up with username, email password and current location, location needs GPS permission from app.

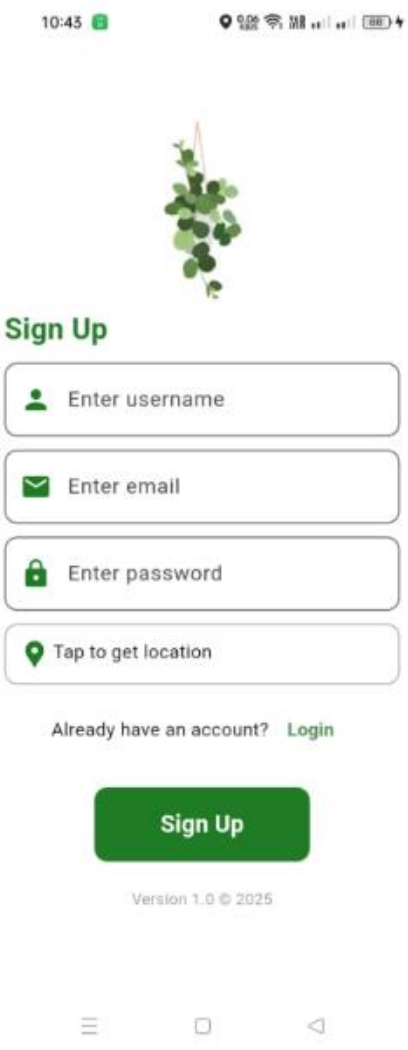

Figure 12: Sign up page

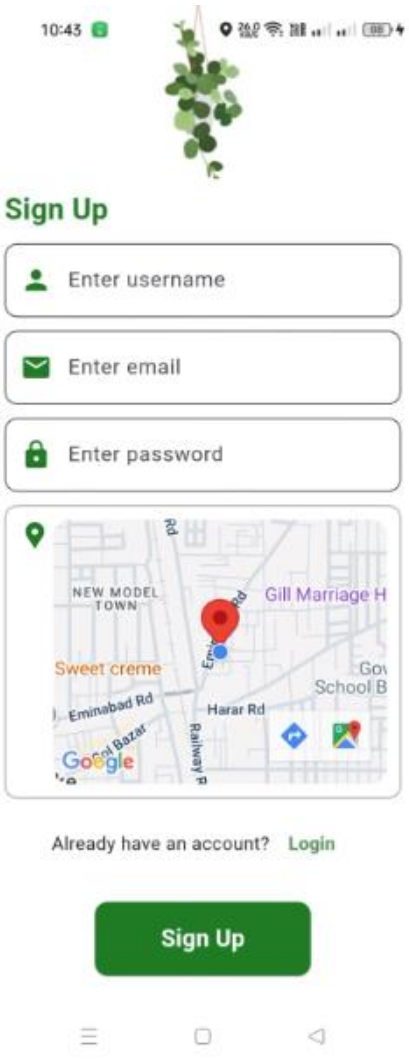

Figure 13: Sign up with location map of user.

- **Login Page:**
  Users can login with email and password, can navigate to sign up if already has account, forget password functionality handled via firebase forgetpassword service.
- **Main Page:**
  Uploaded posts will circulate on main Page, can be filtered by using filters like see all, flowers, sell, give away, swap, will help to get related post.

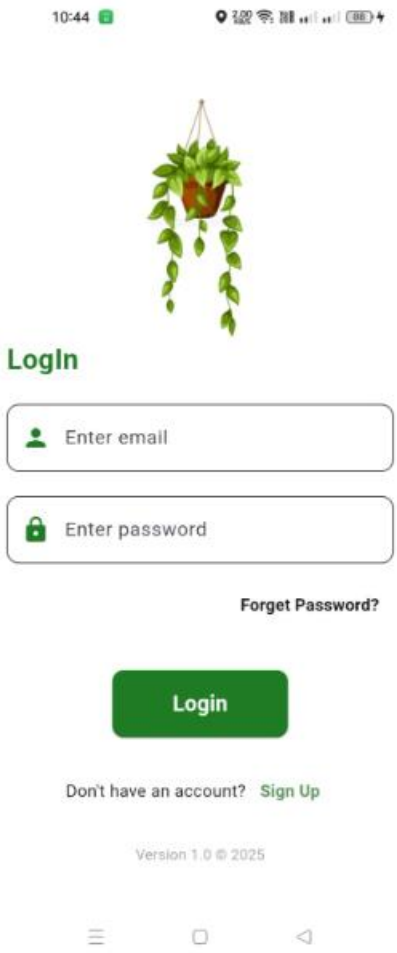

Figure 14: Login page

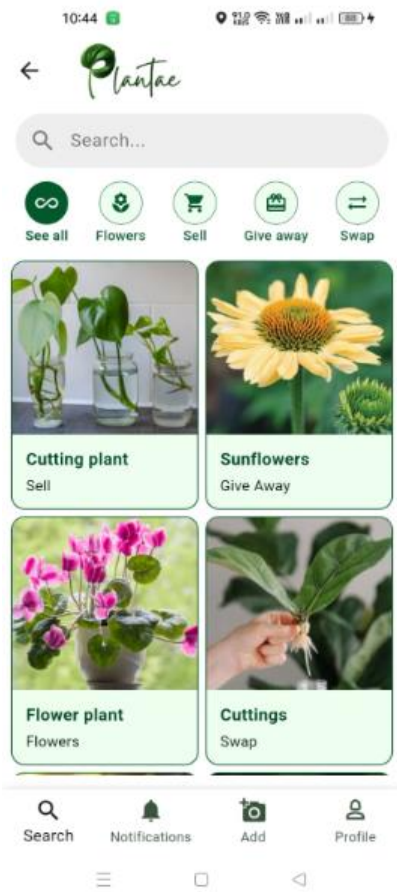

Figure 15: Main page for posts

Figure 16: Filters for search

- **Notifications Page:**
  People who comment, like your post will get notification on this page.
- **Postdetial Page:**
  This page shows the posts detials , including comment section in which people can interact.

Figure 17: Notifications of app

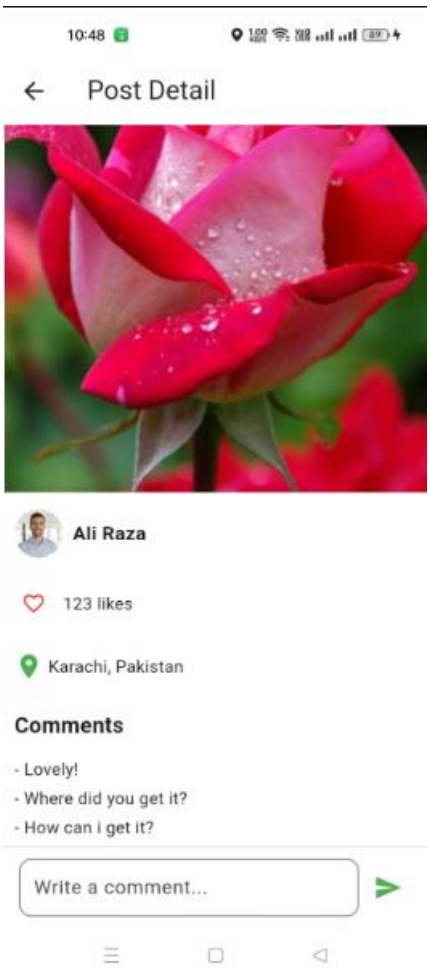

Figure 18: Post detail page

- **Post an ad Page**
  This page takes title, image, description and post type to publish post, save details in firestore with user id of publisher.
- **Account Settings Page**
  This page helps user to manage account detials, user can update password, email, and make account private or public.
- **Edit profile Page**
  User can edit his profile pic, username, about and locaiton with this page.

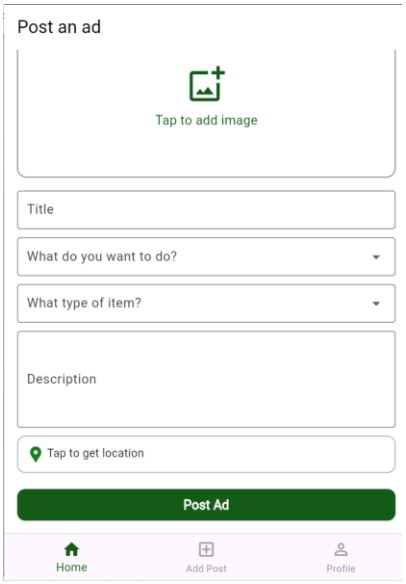

Figure 19: Post an add page

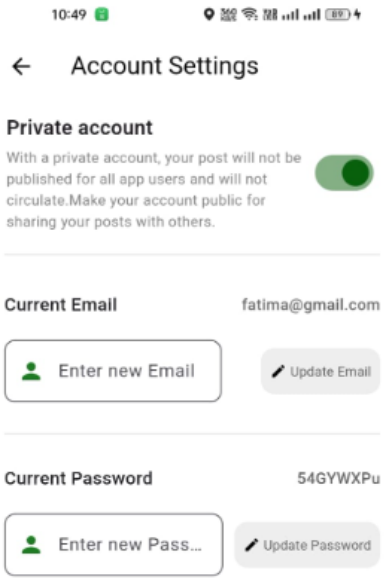

Figure 20: Account settings Page.

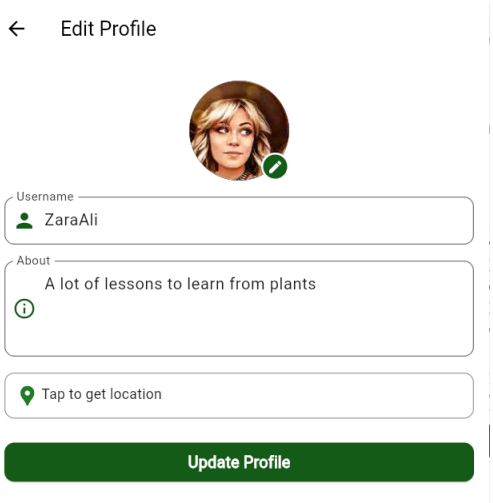

Figure 21: Edit Profile Page.

- **User Profile Page:**
  User profile page with user profile pic, about text, username, and his/her published posts showing, also other section for location map of user , with bottom navigation bar to navigate through app.

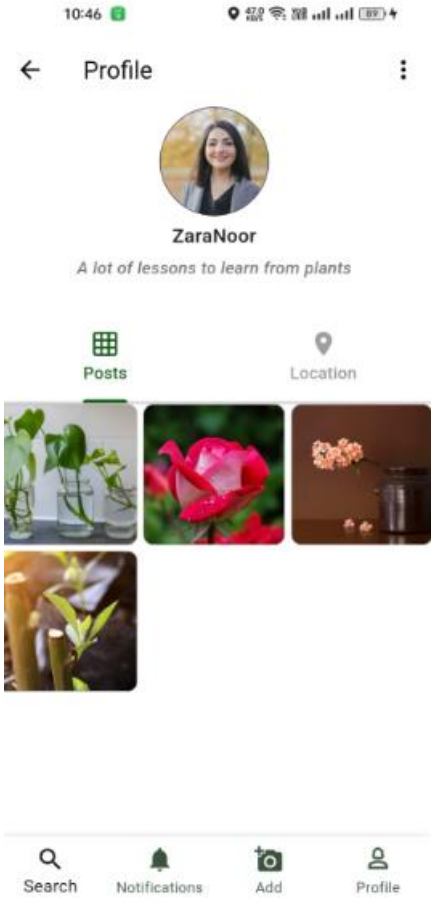

Figure 22: User Profile Page

Figure 23: Location map of user in his profile.

Post Detail Page

Post detail page with all details about post, creator of post, time post has been added and description about post, like button for post to like it.

Comment Section

Each post has a comment section in which user can comment about post, comment shows with profile pic, time duration of comment and text of comment.

Figure 24: Post detail page.

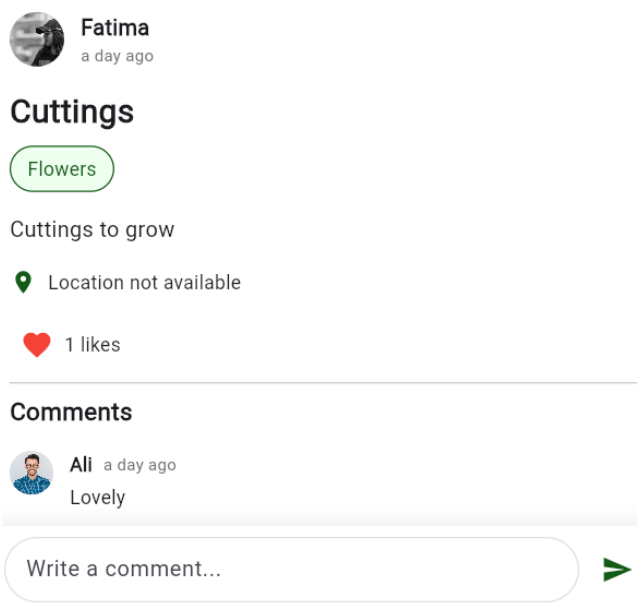

Figure 25: Comment section for post.

1.6 Conclusion

Plantae app is developed for purpose of sharing plants, creating a community of plants lovers and providing them a proper platform. It comprises of all necessary features but need enhancements and further modern features and scabality according to market trends for proper deployment